

Flow Metrics Cheat Sheet

Little's Law | CFD Properties | Stability Checks | Key Principles

Little's Law

$$\text{WIP} = \text{Throughput} \times \text{Cycle Time}$$

John Little, 1961 -- A fundamental law of queueing theory.

If a system is stable, the average number of items in process equals the rate they complete multiplied by the average time each item spends in the system.

WIP

Work In Progress

Items started but not yet finished

Throughput

Completion rate

Items completed per unit of time

Cycle Time

Time in system

How long each item takes from start to done

Unit Conversion Reminder

Throughput per cycle / Sprint length (days) = Throughput per day

Example: 12 items/cycle / 10 days = 1.2 items/day Then check: WIP = 1.2 x 7 = 8.4 (close to actual WIP of 8)

Stable System

WIP = 8 | TH = 12/cycle | CT = 7 days

TH/day = 12/10 = 1.2

Check: 1.2 x 7 = 8.4 ~ 8 = PASS

Unstable System (Overloaded)

WIP = 15 | TH = 7/cycle | CT = 14 days

TH/day = 7/10 = 0.7

Check: 0.7 x 14 = 9.8 != 15 = FAIL

Vacanti's 6 Properties of a CFD

1. Lines must always flow upward (or stay flat) -- never go down
Inspired by: Daniel S. Vacanti - Actionable Agile Metrics for Predictability & When Will It Be Done?
2. The topmost line represents total arrived work.
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